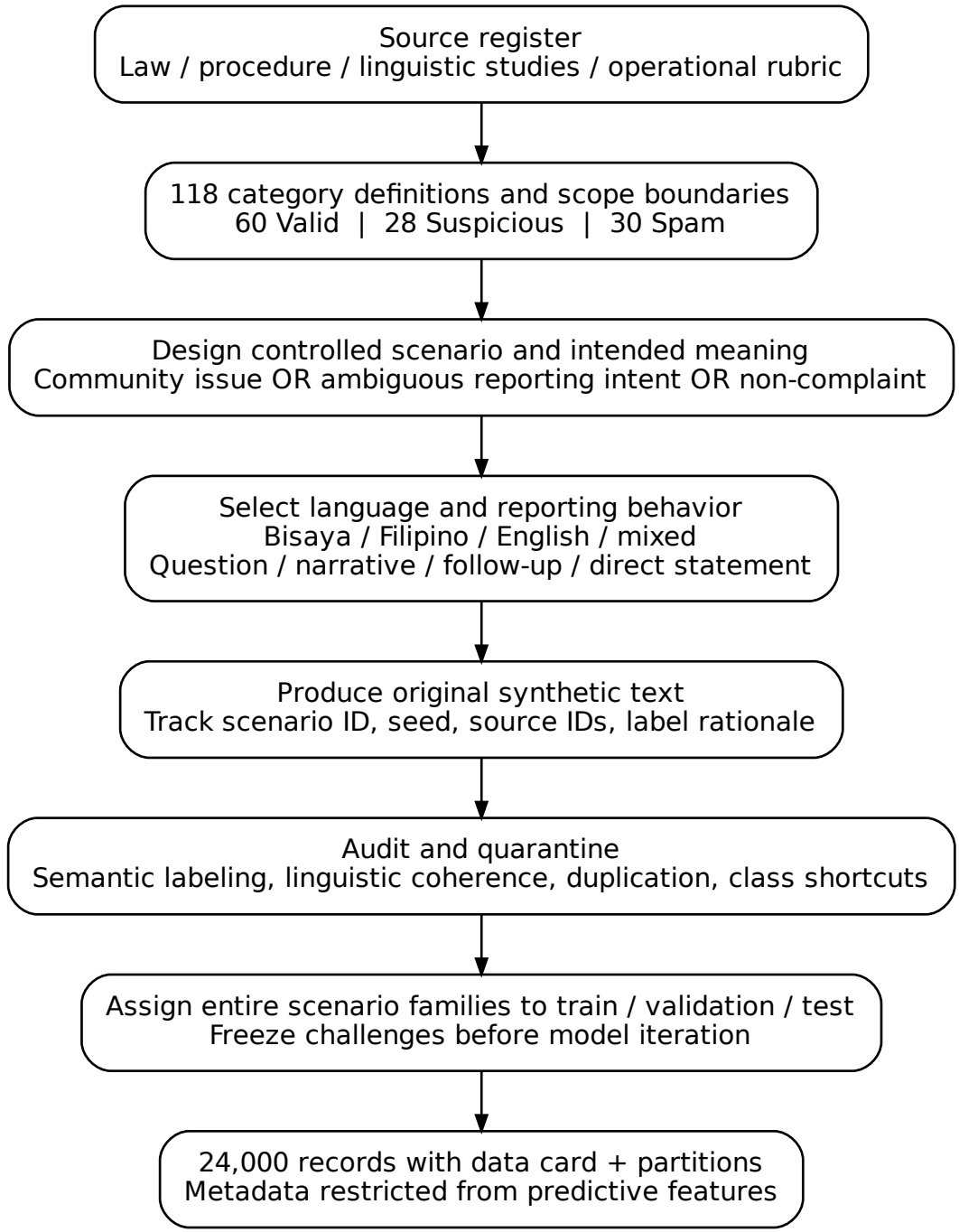RESEARCH-GROUNDED SYNTHETIC DATA WORKFLOW